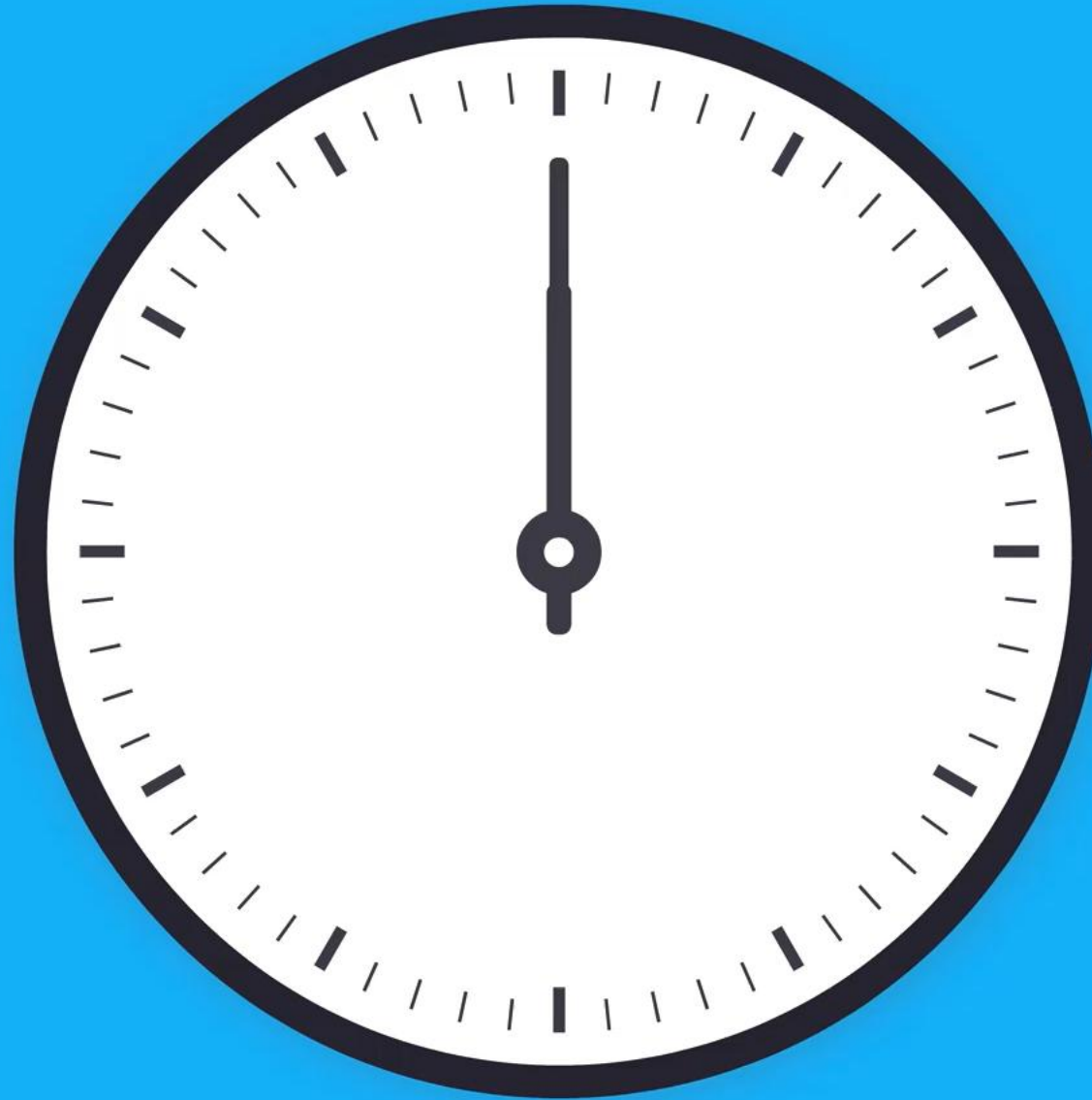
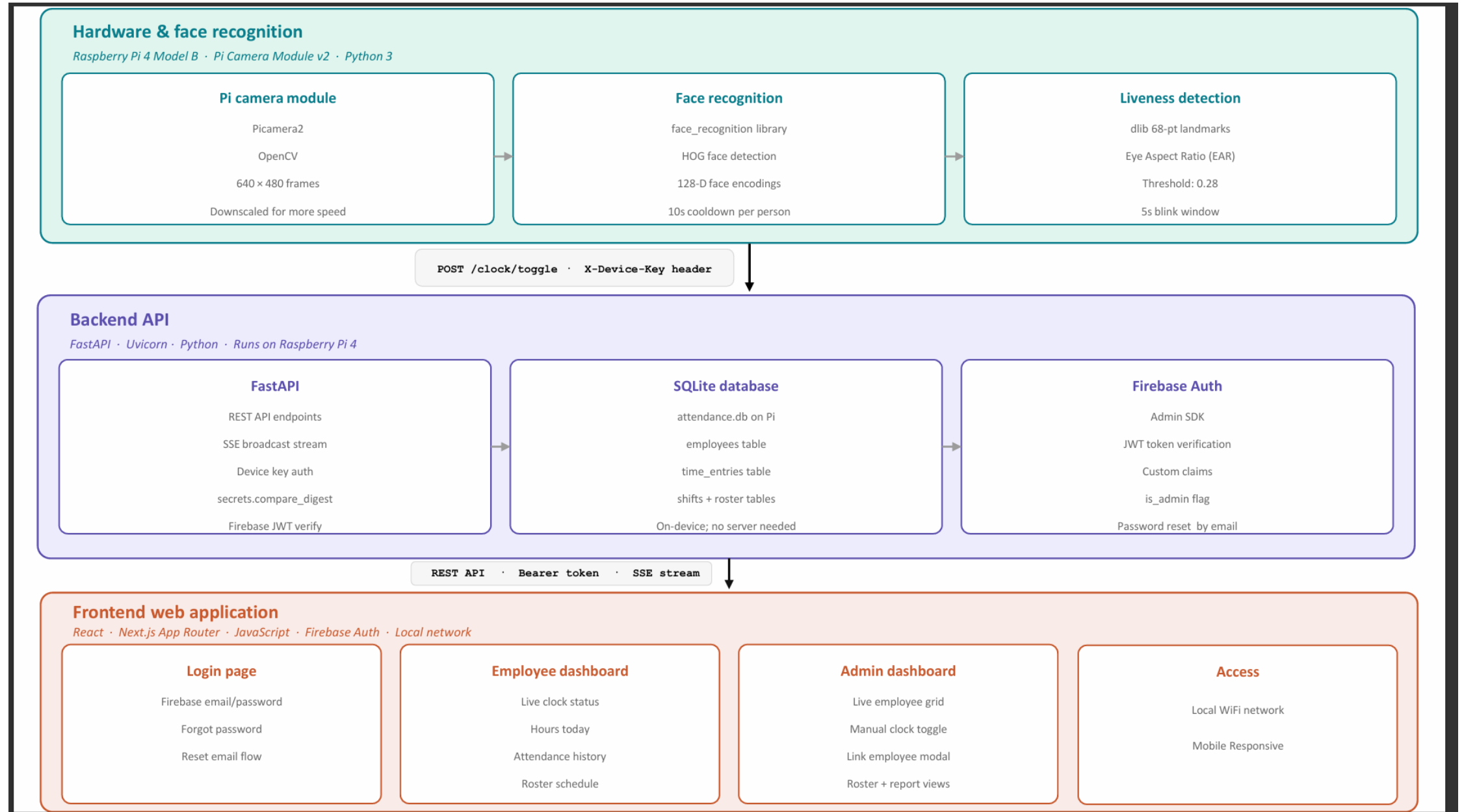


Chuka Orajuba



Smart Punch-In System

Architecture Diagram



Project Overview

The Problem

Manual punch-in systems are susceptible to buddy punching. A contactless, automated solution can help to prevent buddy punching.

The Solution

A Raspberry Pi 4 with a camera module runs real-time face recognition, automatically logging clock-in and clock-out events with no physical interaction required.

The Outcome

100% recognition accuracy in testing. Liveness detection blocked all photo spoofing. Near-instant API response. Full admin and employee web dashboard.

Hardware & Software

Raspberry Pi 4	Raspberry Pi Camera	face_recognition	dlib + OpenCV	FastAPI	SQLite	Firebase Auth	React / Next.js	Server-Sent Events
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Research & Requirements

Background Research

- face_recognition library — dlib 128-D face encodings, HOG detection
- Liveness detection — Eye Aspect Ratio blink analysis
- FastAPI — async Python REST framework with native SSE support
- Firebase Authentication — JWT token verification by Admin SDK
- SQLite — serverless, on the e database for embedded systems
- Next.js App Router — client-side rendered React web application
- Server-Sent Events — one way push

System Requirements

Function

Identify employees in real time, record clock events with timestamp, and confidence score, broadcast live updates to webpage.

Security

Liveness detection to block photo spoofing; device API key for Pi to API auth, Firebase JWT for webpage, constant time key comparison.

Performance

Recognition under 1 second, API response almost instant, SSE latency unnoticeable to users.

Usability

Admin roster, report, and employee linking via webpage, mobile responsive, forgot password.

Implementation & Build Quality

Face Recognition Pipeline

Continuous loop, captures frames from the camera and downscales by 6 before processing. HOG detection is applied, when a face is matched, a 128-dimensional encoding is compared against the known encodings, and the confidence score is stored in the data along with the clock event.

FastAPI Backend

Device, employee, and admin endpoint groups. try/finally on all DB connections. Single connection per request.

Attendance Report

Flags: late_in, early_out, no_show, no_clock_out. Compares first clock-in and last clock-out vs scheduled shift times. 10-min grace period.

Liveness Detection

dlib 68-pt landmarks. EAR averaged across both eyes. Blink confirmed when EAR < 0.28 for 1 or more frames then rises. 5 second window with a countdown on screen.

Shift Roster

Admin creates shift templates and can assign it to selected employees. GET /admin/roster returns full week. Employee can view a 14-day schedule.

Employee Linking

Admin picks unlinked employee from dropdown menu, enters email. Backend creates Firebase account, updates record, sends password invite through REST API.

Problem Solving

FastAPI 405: Route ordering conflict

Issue: `/admin/employees/unlinked` matched by `/employee_id` parameterised route

Fix: Moved all static routes before parameterised routes in `main.py`

CORS 400: OPTIONS error

Issue: `.env` had `CORS_ORIGIN`, while the code had `CORS_ORIGINS`. This prevented me from accessing the webpage on my phone.

Fix: changed the `.env` file to match the code and tested it on the phone to make sure

AI Use

I acknowledge the use of AI in this project to help me with some tasks such as debugging and brainstorming

SQLite database locked

Issue: Two separate connections opened in the same request when trying to create Firebase account and send password email which the database is locked error

Fix: Refactored to single connection per request wrapped in `try/finally` block

EAR threshold: blinks not detected

Issue: Figuring out the threshold for blink detection

Fix: debug logging: open eye 0.39, blink 0.165–0.198 threshold 0.28

Testing

Face Recognition Accuracy

- Tested 20+ times with 100% correct identification in all cases
- Tested at varying distances (0.3 m – 1m from camera)
- Recognition degrades with fewer than 5 training images
- No false positives recorded across all test sessions

API & Backend

- POST /clock/toggle — verified IN/OUT toggle and SSE broadcast
- GET /admin/employees/unlinked — confirmed correct route ordering fix
- POST /admin/employees/link — verified Firebase account creation and invite email
- Database lock fix tested under rapid sequential requests

Liveness Detection

- Photo held in front of camera, clock event correctly blocked each time
- Confirmed blink window timeout fires after 5 seconds with no blink
- EAR threshold calibrated through live debug logging session

Frontend & Integration

- Login to dashboard flow tested on laptop and mobile
- SSE live update verified: clock event on Pi shows an in an instant on the webpage
- Roster assign/clear and shift create/delete tested end to end
- Attendance report flags tested with deliberate late, early, and absent events

Project Management

Sprints

- Sprint 1: Develop, write and submit project proposal
- Sprint 2: Research on project idea and hardware parts that have been ordered
- Sprint 3: Set up a simple facial recognition program and a simple program to count how long a face is clocked in for
- Sprint 4: Setting up a webpage for the project
- Sprint 5: fixing the backend and frontend to make a connection between the two sides
- Sprint 6: Have the basics of the project running and make necessary adjustments
- Sprint 7: Report, poster, video and finishing touches to the project

Weekly Stand-Ups

- Attended weekly supervisor meetings.
- Provided updates with supervisor each week to tell her what I've done and what I'm currently working on.
- Attended the weekly class stand ups
- Plan for following week agreed at each stand up and tell the class what I got done the previous week
- I was also updating the stand ups each week, logging what everyone said each week and checking if the work they done verified what they said they were going to do the week before.
- As well as logging the classes stand ups, I was logging my weekly journal each week, filling in everything I got done every week.

Project Management

Spaces

 **4th Year FYP**  ...









 Summary

 **Timeline**

 Backlog

 Board

 Calendar

 List

 Forms

 Development

 Code

More **4** 

 Search timeline

Epic 

Status category 





Work	Apr / May												May						
	23	24	25	26	27	28	29	30	1	2	3	4	5	6	7	8	9		
Sprints																			
☐ ⚡ FYP-19 Sprint 2																			
☐ > ⚡ FYP-23 Sprint 3																			
☐ > ⚡ FYP-33 Sprint 4																			
☐ > ⚡ FYP-39 Sprint 5																			
☐ > ⚡ FYP-51 Sprint 6																			
☐ > ⚡ FYP-59 Sprint 7																			



Project Management

25-26 6802 -- PROJECT ENGINEERING Notebook

25-26 6802 -- PROJECT ENGINEERING Notebook

+ Add section

+ Add page

Welcome

> _Collaboration Space

> _Content Library

✓ CHUKA ORAJUBA - S...

Teamwork

Weekly Logs

Prompt Logs

Week 12 standup

Week 13 Standup

Week 14 Standup

Week 15 Standup

Week 16 Standup

Week 17 standup

Week 18 Standup

Week 19 Standup

Week 20 Standup

Week 22 Standup

Week 23 Standup

Week 24 Standup

Sean

Last week: finished the poster and made a new architecture diagram for it, also generated a analysis of the yolo performance over the 10 epochs to see how it improved to display it on the poster. and also set up the touch screen to mimic a car infotainment screen.

This week: I want to do the poster, report and video and presentation slides and any minor bugs or changes that may appear along the way

Blockers: None

Johanan

Last Week: I finished the poster and start my report aswell as studying for my mcqs and doing my Fergus and Brian project, I also built the glasses and added in my components.

This week: I plan to finish my report, video, and my presentation.

Blockers: No

Eoin

Last week: I completed my poster, I also started on my report with nearly finishing it, also started reviewing my code to get ready for the presentation.

this week: I am going to finish my report and keep preparing for the presentation

Project Management

25-26 6802 -- PROJECT ENGINEERING Notebook ▾

25-26 6802 -- PROJECT ENGINEERING Notebook



+ Add section

+ Add page



Welcome

Week 13



_Collaboration Space

Week 14



_Content Library

Week 15



CHUKA ORAJUBA - S...

Week 16



Teamwork

Week 17



Weekly Logs

Week 18



Prompt Logs

Week 19

Week 20

Week 21

Week 22

Week 23

Week 24

Week 24

21 April 2026 10:22

This week I am continuing on with my report and I hope to have it finished by Friday at the latest then I will make a start on doing my presentation and video for the demo next week.

I have finished my report and read over it to make sure I didn't make any silly errors or spelling mistakes. Once I had that done I started on my presentation and got it done and gave it a read through to make sure everything is fine. I also done a practice presentation to get myself familiar with presenting in front of the lecturers.

I then got started on the video and made sure I covered everything necessary in the 2 minutes and once I had it done I done the project summary form

Github Links

- Backend: <https://github.com/chuka111/FYP.git>
- Frontend: https://github.com/chuka111/FYP_react.git